The empirical recurrence for OEIS A204565, proved by counting patterns

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Abstract

OEIS A204565 counts arrays subject to a local condition *and* to the clause "new values 0..3 introduced in row major order", which says the array is written in canonical form. That clause is not local, so a transfer matrix over the alphabet does not see it. It does not have to: what is being counted is not arrays but equality patterns, and the number of patterns is recovered from the numbers L_i of arrays over an i-letter alphabet by inverting a falling factorial. The result is the clean identity $4!\,a(n) = \sum_i \binom{4}{i} D_{4-i} L_i(n)$ with D_m the derangement numbers, and each L_i is an ordinary walk count. A second reduction — the chain is strongly lumpable over the patterns, because the condition is relabelling-invariant — brings the state space down from 30 to $S = 7$. The entry's empirical recurrence of order 3, contributed by R. H. Hardin, then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A204565 is "Number of (n+1) X 2 0..3 arrays with every 2 X 2 subblock having equal diagonal elements or equal antidiagonal elements, and new values 0..3 introduced in row major order.". It has offset 1 and begins

$$8, 42, 256, 1682, 11448, 79162, 551216, 3849762, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 11*a(n-1) - 31*a(n-2) + 21*a(n-3).

As of the "Last modified" line on the live entry (Jun 07 2018 11:47:13, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 11\,a(n-1) - 31\,a(n-2) + 21\,a(n-3) \tag{1}$$

for all $n > 2$.

2 What the canonical-form clause counts

Fix the shape: 2 columns and $L = n + 1$ rows, so $L \cdot 2$ cells. Call a partition of the cells into nonempty classes a *pattern*; an array over an alphabet determines a pattern, two cells lying in the same class exactly when they carry the same value.

Every 2×2 subblock of the array occupies two consecutive rows and two consecutive columns; writing those two rows as r and s the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix},$$

and the entry's requirement on it is

$$\alpha = \delta \quad \text{or} \quad \beta = \gamma. \tag{2}$$

The condition is stated purely in terms of equalities between entries, so it is invariant under any relabelling of the values and is therefore a property of the pattern alone. Call a pattern *admissible* when it has the property.

The clause "new values 0..3 introduced in row major order" selects, from each class of arrays that differ only by a relabelling, exactly one representative: reading the cells in row major order, the first occurrence of the value v must precede the first occurrence of $v+1$. Every pattern with at most $K = 4$ classes has exactly one such representative, and every array in canonical form is the representative of its own pattern. Hence, writing $N_j(n)$ for the number of admissible patterns with exactly j classes,

$$a(n) \;=\; \sum_{j=0}^{4} N_j(n). \tag{3}$$

Lemma 1. *Let $L_i(n)$ be the number of arrays over an alphabet of i letters satisfying the local condition, with no canonical-form clause. Then*

$$L_i(n) = \sum_j N_j(n)\, i(i-1)\cdots(i-j+1), \qquad 4!\sum_{j=0}^{4} N_j(n) = \sum_{i=0}^{4}\binom{4}{i} D_{4-i}\, L_i(n),$$

where $D_m = m!\sum_{t=0}^{m}(-1)^t/t!$ is the number of derangements of m letters.

Proof. An array over i letters satisfying the condition determines an admissible pattern together with an injection of its classes into the alphabet, and conversely; a pattern with j classes admits $i(i-1)\cdots(i-j+1)$ injections. That is the first identity. Writing $i(i-1)\cdots(i-j+1) = j!\binom{i}{j}$ it reads $L_i = \sum_j (j!N_j)\binom{i}{j}$, so by binomial inversion $j!N_j = \sum_i (-1)^{j-i}\binom{j}{i}L_i$. Summing over $j \le 4$ and exchanging the order of summation,

$$\sum_{j=0}^{4} N_j = \sum_{i=0}^{4} L_i \sum_{j=i}^{4} \frac{(-1)^{j-i}}{i!\,(j-i)!} = \sum_{i=0}^{4}\frac{L_i}{i!}\sum_{t=0}^{4-i}\frac{(-1)^t}{t!},$$

and multiplying by $4!$ turns the coefficient of L_i into $\frac{4!}{i!}\sum_{t\le 4-i}(-1)^t/t! = \binom{4}{i}(4-i)!\sum_{t\le 4-i}(-1)^t/t! = \binom{4}{i}D_{4-i}$, an integer. $\square$

3 Each L_i is a walk count, and the chain lumps

Fix i and let $V_i = \{0,\dots,i-1\}^2$ be the possible rows. For $r, s \in V_i$ declare $r \to s$ an edge when placing s directly after r satisfies (2) at every column, and let M_i be the adjacency matrix. An array is exactly a sequence of rows, and its condition is exactly the conjunction of the edge conditions along that sequence, so

$$L_i(n) \;=\; \mathbf{s}_i^\top M_i^{\,n}\mathbf{1}.$$

Because the condition involves no row before the first, every row may start a walk, and $\mathbf{s}_i = \mathbf{1}$.

Lemma 2. *Let $\pi(r)$ be the equality pattern of the row r. For all r, r' with $\pi(r) = \pi(r')$ and every pattern Q,*

$$\#\{s :\ r \to s,\ \pi(s) = Q\} \;=\; \#\{s :\ r' \to s,\ \pi(s) = Q\}\,.$$

Consequently the numbers $\widehat{M_i}[P][Q]$ obtained by taking any representative of P are well defined, and with $e_P = i(i-1)\cdots(i-|P|+1)$ the number of rows of pattern P,

$$\mathbf{s}_i^\top M_i^{\,m}\mathbf{1} \;=\; \sum_P \varepsilon_P\, f_m(P), \qquad f_0 \equiv 1, \quad f_{m+1}(P) = \sum_Q \widehat{M_i}[P][Q] f_m(Q),$$

where $\varepsilon_P = e_P$ if rows of pattern P may start a walk and 0 otherwise.

Proof. If $\pi(r) = \pi(r')$ then the map sending the value of r at a column to the value of r' at that column is a well-defined injection between the letters used, and extends to a permutation σ of the alphabet with $r' = \sigma(r)$. The edge condition is a statement about equalities among entries, so $r \to s$ if and only if $\sigma(r) \to \sigma(s)$; and $\pi(\sigma(s)) = \pi(s)$. Hence $s \mapsto \sigma(s)$ is a bijection between the two successor sets preserving patterns, which is the first claim. It says the chain is strongly lumpable over π, so $\big(M_i^m\mathbf{1}\big)(r)$ depends on r only through $\pi(r)$; calling that value $f_m(\pi(r))$ gives the recursion, and summing over r in each class gives the displayed identity, the number of rows of pattern P being the number of injections of its $|P|$ classes into the alphabet. $\square$

Assembling the 4 lumped chains into one block-diagonal matrix N of size $S = 7$ and putting the weights of Lemma 1 into the start vector, $\mathbf{v} = \big(\binom{4}{i} D_{4-i}\varepsilon_P\big)_{i,P}$, gives

$$4!\; a(n) \;=\; \mathbf{v}^\top N^{\,n}\mathbf{1}. \tag{4}$$

The unlumped alphabet had 30 rows in all; the lumped chain has $S = 7$ states.

4 The criterion, and the computation

Let $q(t) = t^3 - \sum_i c_i\, t^{3-i}$ be the characteristic polynomial of (1).

Lemma 3. *For every n with $m = n + 1 - 1 \geq 3$,*

$$4!\Big(a(n) - \sum_i c_i\, a(n-i)\Big) \;=\; \mathbf{v}^\top N^{\,m-3} q(N)\mathbf{1}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top N^j q(N)\mathbf{1} = 0$ for every $j \geq 0$, and it is enough to check $j = 0, 1, \ldots, S-1$.

Proof. Substitute (4) into the left side and factor N^{m-3} out of $N^m - \sum_i c_i N^{m-i}$. The scalar $4!$ is nonzero, so it does not affect whether the left side vanishes. For the last clause put $w = q(N)\mathbf{1}$: by Cayley–Hamilton N^S is an integer combination of $I, N, \ldots, N^{S-1}$, so each $\mathbf{v}^\top N^j w$ with $j \geq S$ is the corresponding combination of earlier values. $\square$

Theorem 4. *The recurrence (1) holds for every $n > 2$, which is the statement of Section 1.*

Proof. The lumped transition counts were obtained by enumerating, for one representative row of each pattern, all admissible successors and sorting them by pattern — the successors being generated column by column, since the edge condition is a conjunction of one constraint per column. The vector $w = q(N)\mathbf{1}$ was then formed by 3 matrix–vector products in exact integer arithmetic and $\mathbf{v}^\top N^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 2 + 1$ onwards, and the run continued until w was the zero vector, which settles all larger j at once. Lemma 3 gives the theorem. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry's DATA: the right-hand side of (4), divided by 4!, was evaluated at every one of the 20 published indices and agrees exactly. This is a strong test of the modelling and not only of the arithmetic — the inversion of Lemma 1, the reading of the local condition, and the alignment of the entry's index with the number of rows would each show up as a mismatch, and the quotient by 4! came out an integer at every index.

Second, the lumped chain was checked against the unlumped one: for the small shapes the walk counts $\mathbf{s}_i^\top M_i^m \mathbf{1}$ were computed directly over the full alphabet as well and agree with $\sum_P \varepsilon_P f_m(P)$ term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 3 was run until the working vector was identically zero rather than to a sampled prefix.

References

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